A Software-based Mitigation against Inaudible Voice Attacks

Guided by
Dr. Anita John

Presented by
Shreya Veeraraghav
S7 CSE – Gamma
52

Title

Learning normality is enough: a software-based mitigation against inaudible voice attacks.

Authors

Xinfeng Li, Xiaoyu Ji, Chen Yan, Chaohao Li, Yichen Li, Zhenning Zhang, and Wenyuan Xu.

Published in

Proceedings of the 32nd USENIX Conference on Security Symposium (SEC '23). USENIX Association, USA, Article No. 138, pages 2455–2472.

Overview

1 Introduction

2 Motivation

3 Objective

4 Inaudible Voice Attacks

5 Attack Investigation

6 System Overview

7 Dataset

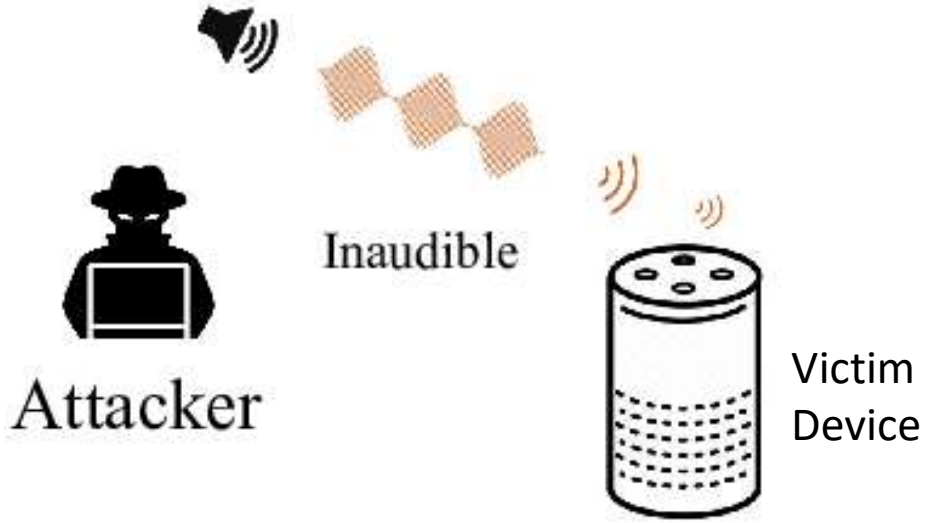
8 Overall Performance

9 Comparison with existing defenses

10 Conclusion

11 References

INTRODUCTION



- sending commands to voice-activated systems at frequencies beyond the range of human hearing
- manipulate devices that are programmed to recognize voice commands without the user being aware of it

- **Victim Devices** : Voice – activated systems (smartphones, smart speakers)
- **DolphinAttack** first demonstrated in 2017

MOTIVATION

- smartphone companies indicate software mitigations that can instantly protect miscellaneous existing devices without any hardware modification are in need
- existing devices designed with diverse features and resources - need a defense that is transferable and lightweight.
- patterns of inaudible voice attacks are diverse, benign audios share similar patterns - motivated to detect the attacks (the anomaly) by learning the benign audios (the normality).

OBJECTIVE

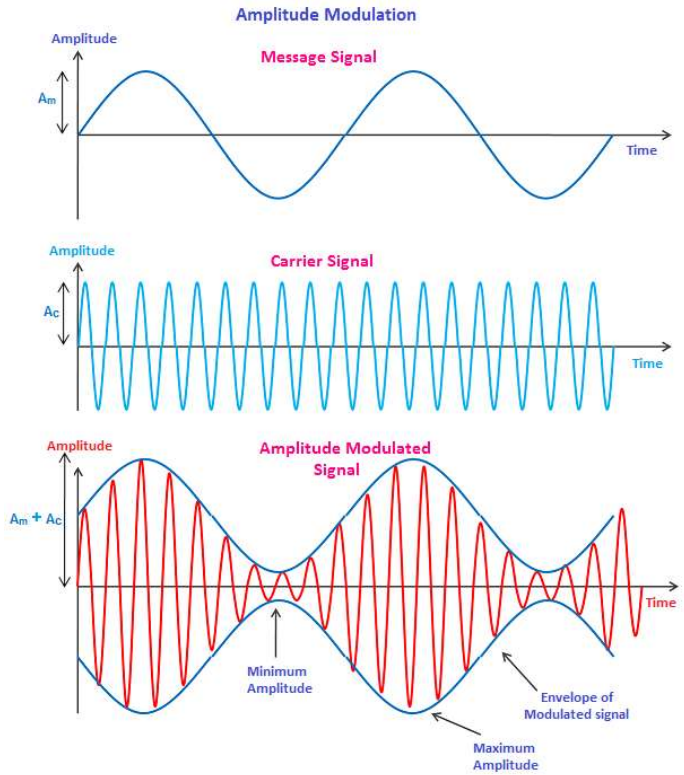
A holistic unsupervised learning algorithm that can detect inaudible voice attack for many smart devices only based on accessible normal audios, that meets the following goals :

1. Lightweight
2. Universal
3. Unsupervised

INAUDIBLE VOICE ATTACKS

1. Modulation : Attackers modulate a voice command to an ultrasonic frequency, by **amplitude modulation** (AM).

2. Device Reception : Voice recognition systems detect them because their microphones have wider frequency responses than the human ear.





Voice assistant

3. Command Execution : device interprets the modulated signal as a legitimate voice command and performs the associated action.

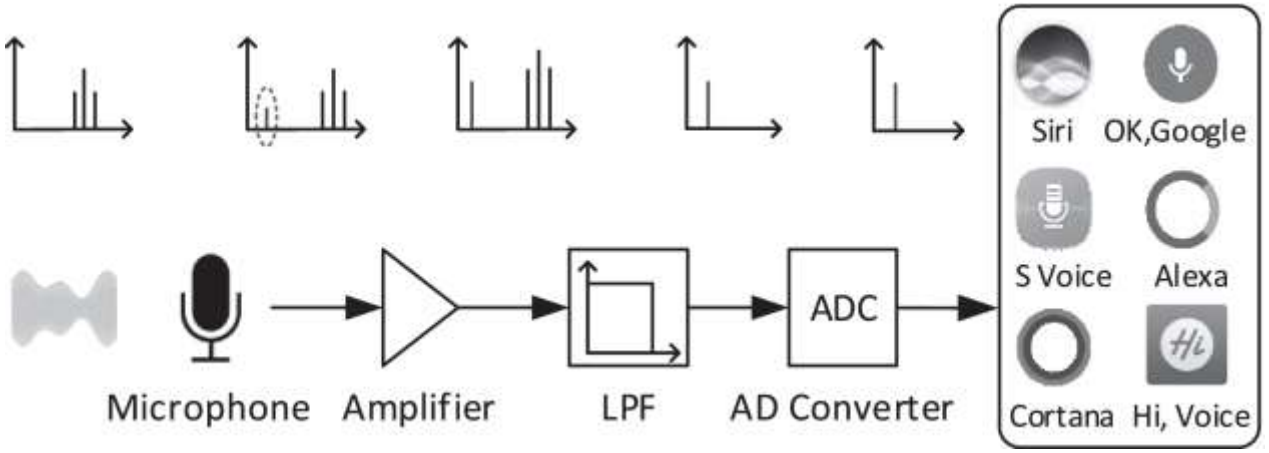


Fig. Workflow of an Inaudible Voice Attack

4. Attack Distance : attacker only needs to be within a certain distance (a few meters) from the target device to perform the attack.

ATTACK INVESTIGATION

RQ 1: What are the differences between benign audio and inaudible voice attacks?

RQ 2: How different are the benign audio and inaudible voice attacks?

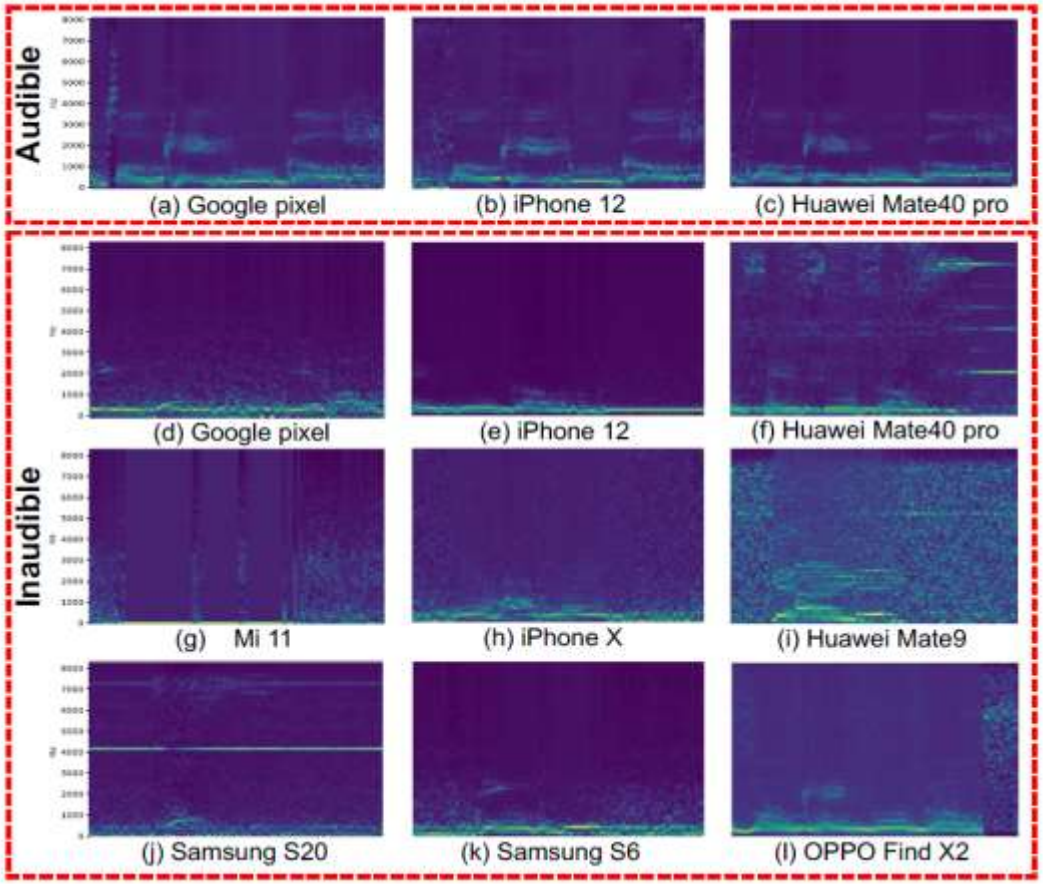
RQ 3: Why do the attack patterns vary on different target devices?

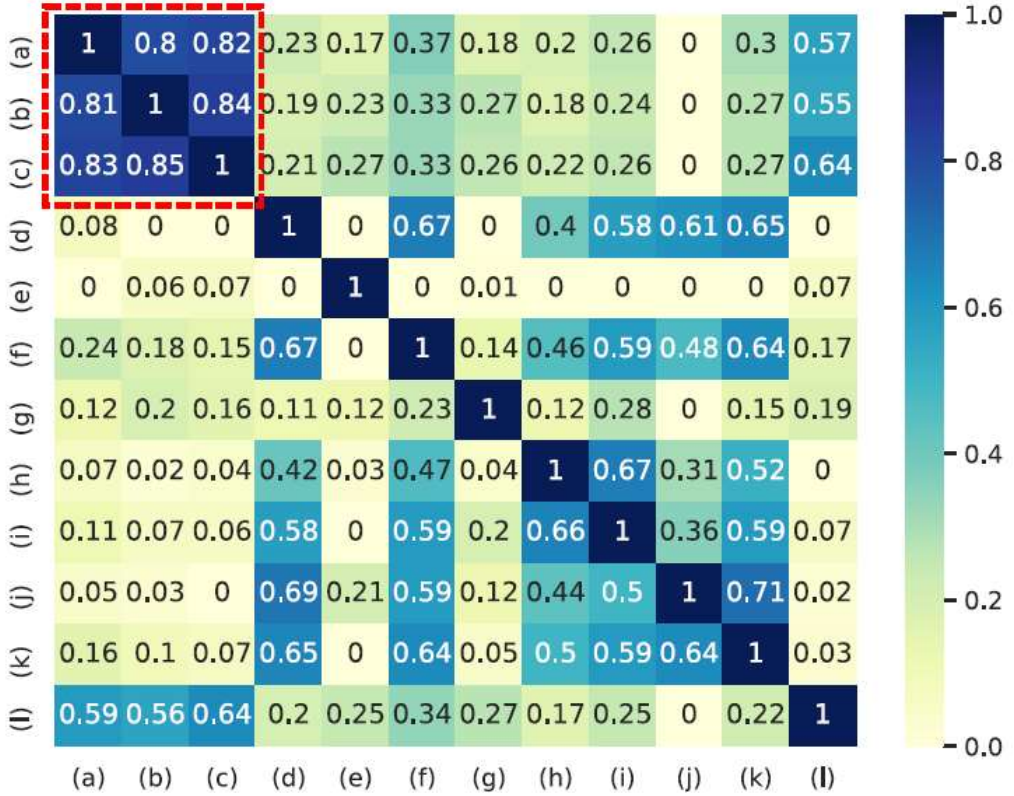
RQ1 : What are the differences between benign audio and inaudible voice attacks?

- quality of the injected attack audio varies greatly with the target device
- quality of the recorded benign voices is clear and similar across all devices
- attack power directly affects the audio quality
- speech commands have little impact on the attack effectiveness

RQ2 : How different are the benign audio and inaudible voice attacks?

- spectrums of the 3 audible cases are quite similar
- Patterns of inaudible commands are different from each other

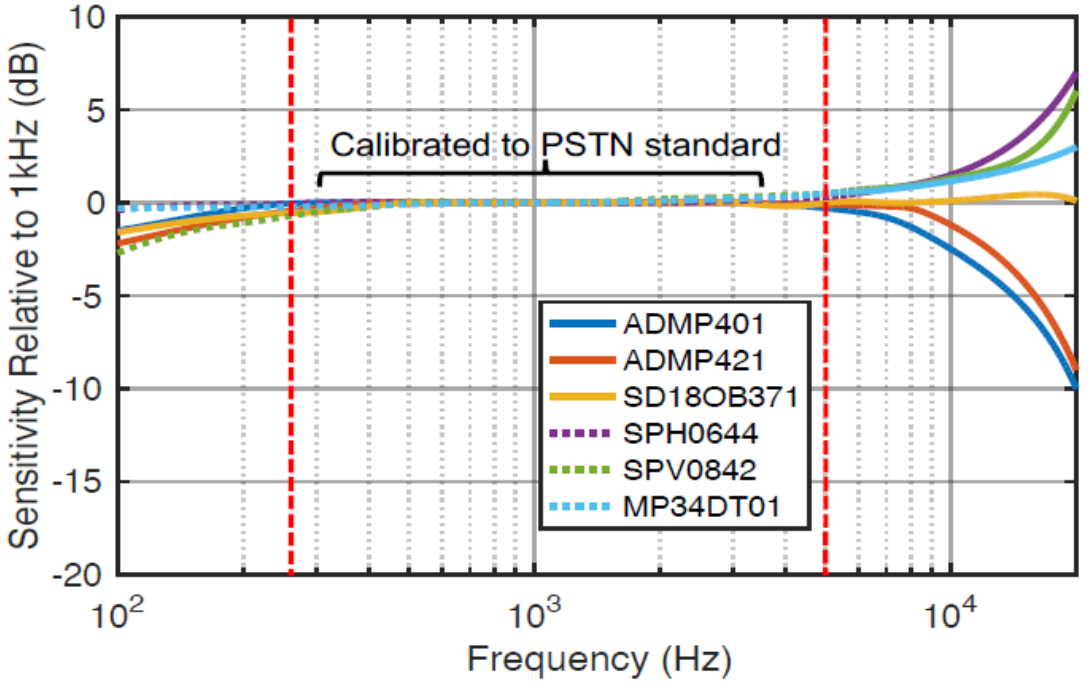




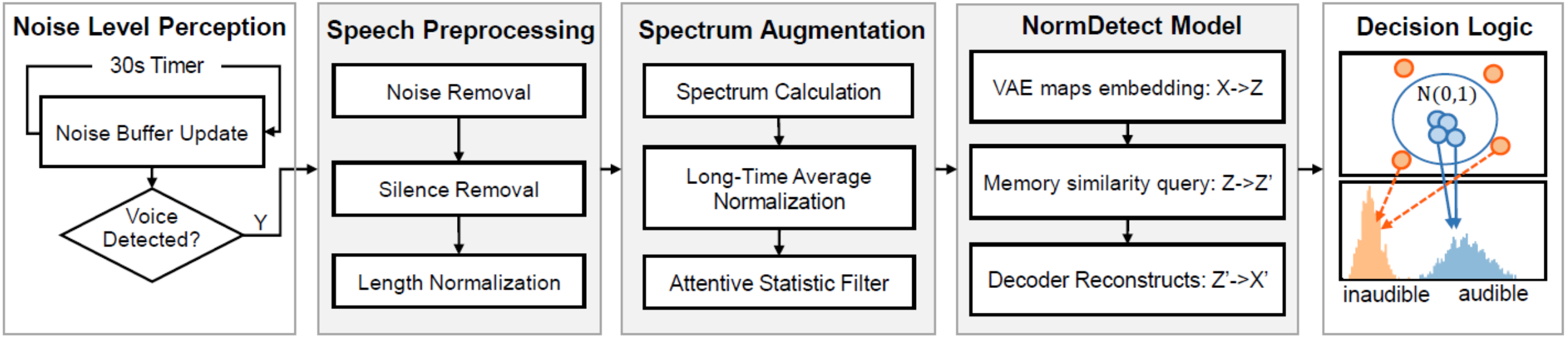
RQ3 : Why do the attack patterns vary on different target devices?

- Root cause : diverse frequency response of microphones in the upper 10 kHz band

- normal voices more similar to each other because the microphone's frequency response from 300 to 3400 Hz is close due to the manufacturers' calibration to comply with the PSTN standard.



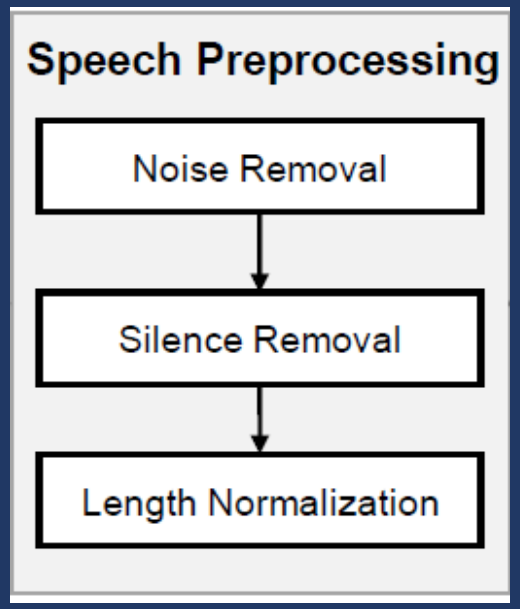
SYSTEM OVERVIEW



3 key stages:

1. Speech preprocessing
2. Spectrum Augmentation
3. NormDetect Model

Speech Preprocessing



1. Noise Removal

   - since the ambient noise is somewhat similar to the abnormal microphone response caused by ultrasonic waves

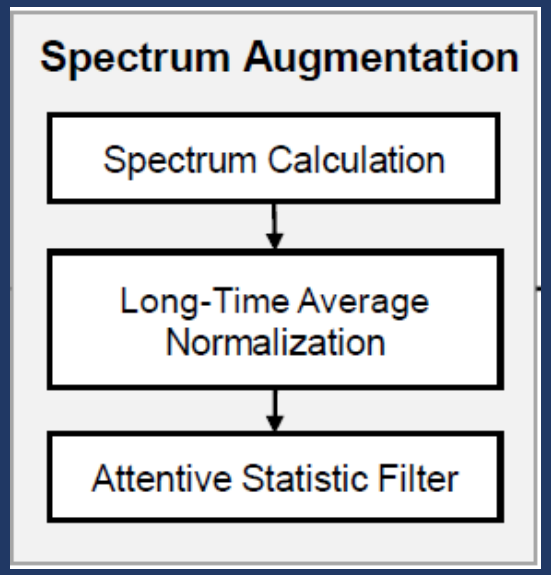
2. Silence Removal

   - ensures that the model can focus on learning voiced features

3. Length Normalization

   - ensures that all voice commands, regardless of their original duration, have a consistent input length when fed into the machine learning model.

Spectrum Augmentation



1. Spectrum Calculation/Generation

   - generates the spectrum using Short-Time Fourier Transform(STFT)

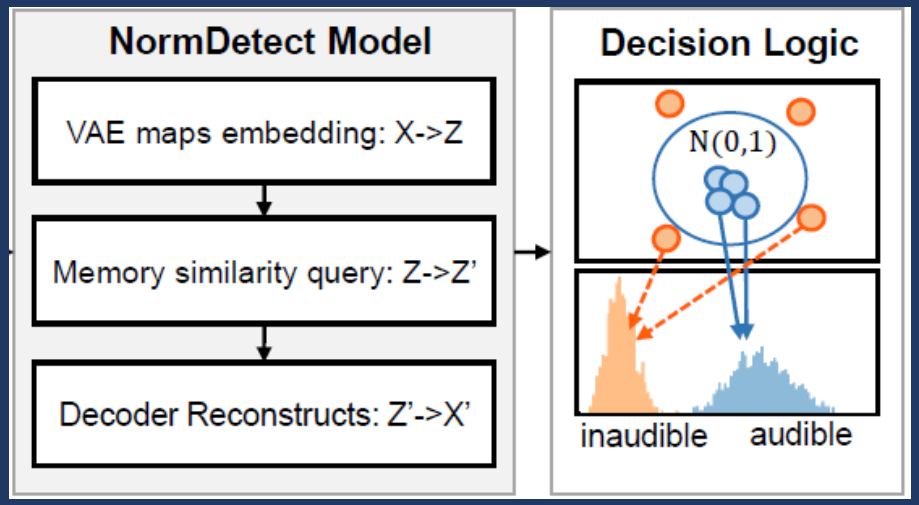
2. Long-Time Average Normalization

   - generates Long-Time Average Spectrum(LTAS) to minimize variations of different speech content and speaker

3. Attentive Statistical Filter

   - enhances high discriminative information between audible and inaudible classes using the Fratio method.

NormDetect Model



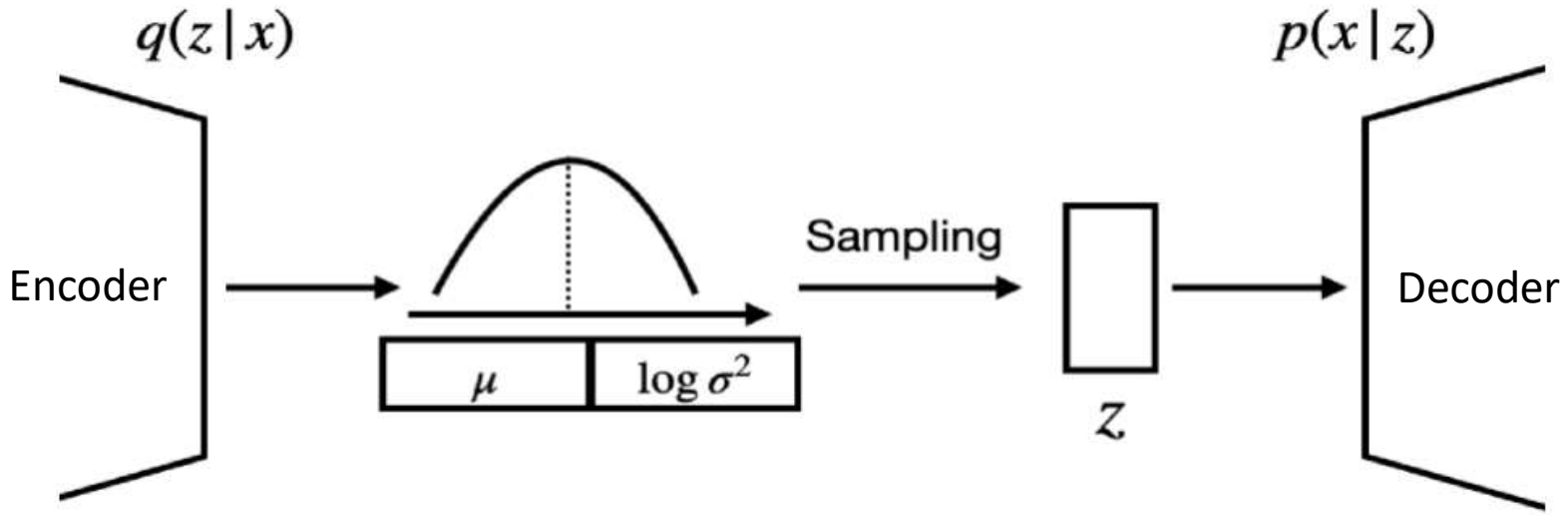
3 stages :

1. maps the augmented spectrums into a latent space (X -> Z)
2. queries the memory module that only stores normal embeddings to make each input similar to normal patterns (Z -> Z')
3. reconstructs the embedding to a spectrum (Z' -> X')

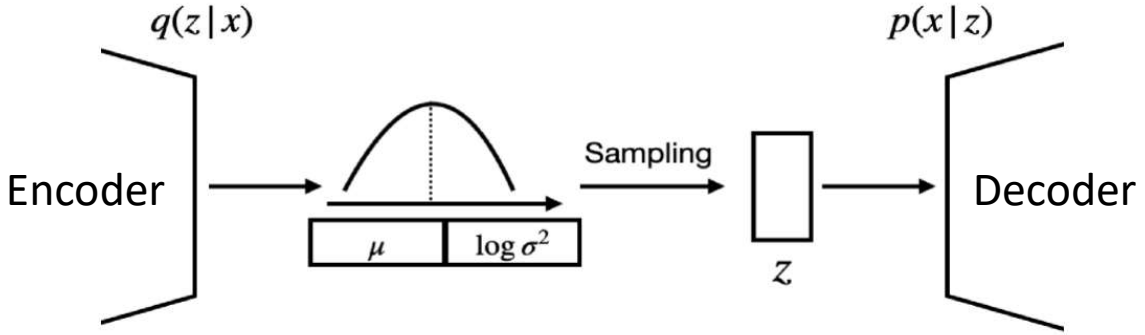
- Uses a Variational Autoencoder

What is a Variational Autoencoder (VAE) ?

Variational autoencoders (VAEs) are generative models that learn compressed representations of their training data as *probability distributions*, which are used to generate new sample data by creating *variations* of those learned representations.



Structure of a VAE



Encoder :

- maps input x to a Gaussian distribution over the latent space
- learns 2 variables – mean and variance, which define the Gaussian distribution for each input

Decoder :

- latent variable z is sampled from this Gaussian distribution,

which is used by the decoder to reconstruct the input or generate $z \sim \mathcal{N}(\mu(x), \sigma^2(x))$

$$z = \mu + \sigma \cdot \epsilon$$

Loss Function

$$L_{total} = L_{recon} + \beta L_{KL}$$

The training of the VAE is guided by two main loss components:

a. **Reconstruction Loss** (L_{recon}) :

- measures how well the decoder can reconstruct the original input **x** from the latent variable **z**.

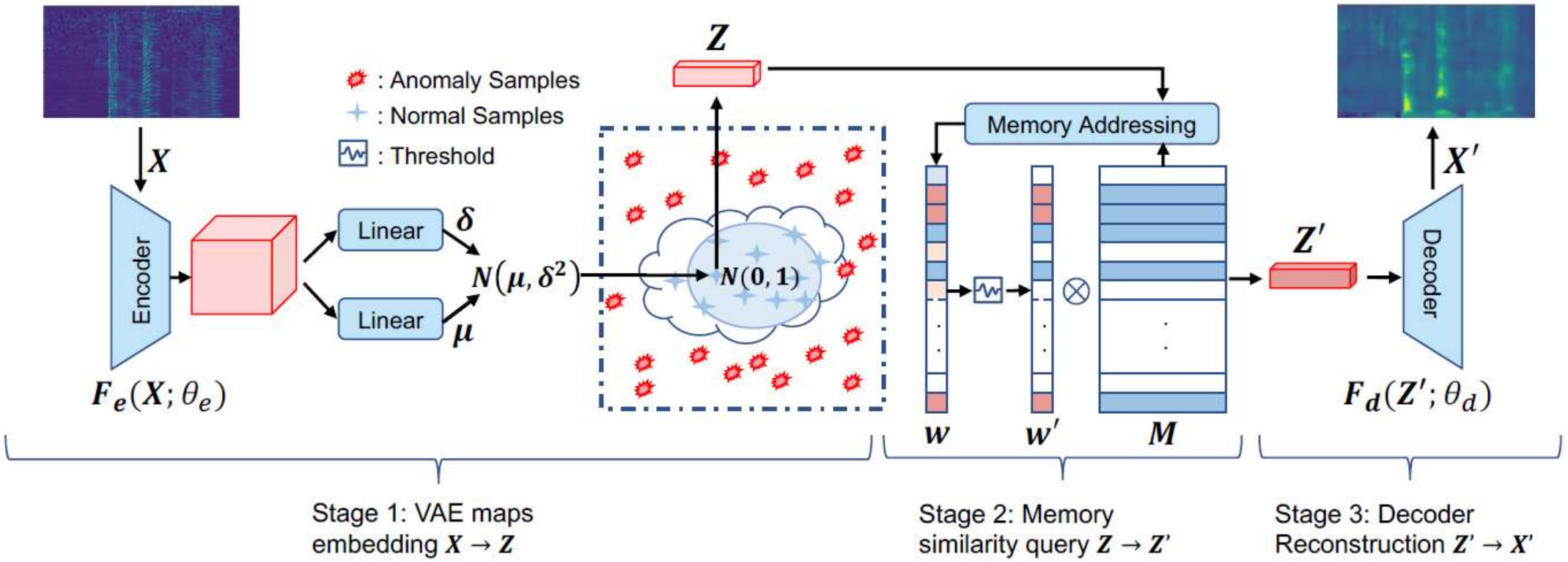
b. **KL(Kullback-Leibler** $L_{recon} = -\sum(x \cdot \log(\hat{x}) + (1-x) \cdot \log(1-\hat{x}))$ (Cross Entropy)

- to measure how much the learned latent di L_{KL} tion differs from prior distribution.

$$L_{KL} = D_{KL}\left[q_\phi(z|x) \parallel p_\theta(z)\right] = \int q_\phi(z|x) \log\left(\frac{q_\phi(z|x)}{p_\theta(z)}\right) dz$$

NormDetect Model

Principle : the model reconstructs the normal well based on standard training data while reconstructs poor for abnormal data.



STAGE 1: VAE maps embedding X -> Z

2 goals –

- Successful reconstruction
- Constraint embedding

- Maps the input spectrum X into a latent embedding Z following Gaussian distribution
- Reparameterization technique used to add noise so that VAE trains to reconstruct various normal patterns in a robust manner

STAGE 2: Memory Similarity Query Z -> Z'

Objective : to constrain the generalization capability of VAE to keep the inaudible voice commands from being falsely accepted

- Memory module consists of items mi which represent patterns of normal audio commands

$$w = ZM = [Zm_1, Zm_2, \ldots, Zm_n]$$

$$Z' = w'M = \sum_{i=1}^{N} \omega_i' m_i$$

where,

w – similarity weights vector

w' – refined weights vector

Z' – adjusted embedding

STAGE 3: Decoder reconstructs Z' -> X'

- Reconstructs spectrum X' from queried embedding Z'
- Evaluated training using Cross Entropy
- Anomaly score computed using negative log-likelihood between X and X'

$$\mathrm{NLL} = -\log(P(X|X'))$$

DATASET

Table 1: The details of two datasets in our experiment

	Speakers	Data Source	Number of Utterance	Language	Average Duration
Training Set	97	Fluent Speech Commands Dataset[9]	30,043	English	3.16s
Test Set	28	Audible & Inaudible Voice Commands Corpus	383,320	English & Chinese	1.31s

Table 6: List of Audible & Inaudible Voice Commands Dataset

Developer	English Wake-up Words	Developer	Chinese Wake-up Words	Phrases	Command
Amazon	Echo	JD	DingDongDingDong	serious everywhere	Take the picture
Amazon	Computer	MeiZu	NiHaoMeiZu	everyone hundred	Call my mom
Amazon	Amazon	Tencent	XiaoWeiXiaoWei	period anywhere	Navigate to my home
Google	OK Google	Lenovo	NiHaoLianXiang	unfortunately	Restart phone now
Google	Hey Google	Huawei	XiaoYiXiaoYi	together difficult	Turn off the light
Microsoft	Hey Cortana	Huawei	NiHaoYoYo	didn't agreement	
Samsung	Hey Bixby	Mobvoi	NiHaoWenWen	immediately connect	
Apple	Hey Siri	OPPO	XiaoBuXiaoBu		
OPPO	Hey Breeno	OPPO	NiHaoXiaoBu		

OVERALL PERFORMANCE

Evaluation Metrics

- False Rejection Rate (FRR) :
  - rate at which the system falsely rejects audible commands
- False Acceptance Rate (FAR) :
  - rate at which an inaudible voice command is wrongly accepted by the system as normal.

- A higher AUC(Area Under Curve) score indicates that the system has more discriminative power against inaudible voice attacks.

OVERALL PERFORMANCE

Result

The model demonstrates :

- Strong overall discriminatory power
- Adaptability across devices
- Real-World Applicability

Evaluation Setting.	AUC.
All Data Mixed	99.11%
All Data Averaged	99.48%

COMPARISON OF EXISTING DEFENSES

NAME	HARDWARE INDEPENDENT	ATTACK SAMPLE INDEPENDENT	SAMPLES	DEVICE	METHODOLOGY
Zhang et al.	✓	✗	24	1 device	SVM
LipRead	✓	✗	2368	3 devices	Elliptical Classifier
GuardSignal	✗	✓	/	1 prototype	Signal Fingerprinting, Acoustic Feature Extraction
EarArray	✗	✗	13000	1 prototype mic array	Beamforming and Microphone Array Processing
Li et al.	✗	✗	1560	3 mic arrays	Convolutional Neural Network
NormDetect	✓	✓	383320	24 devices	VAE with Memory Similarity Query

CONCLUSION

- discovered that patterns of inaudible voice attacks are diverse and distinct from that of the audible, while audible voices share similar patterns
- designed a universal detection model NormDetect, different from the supervised learning approach of existing studies
- NormDetect performs an average AUC of 99.48% , suggesting its effectiveness in detecting inaudible voice attacks.

References

1. Li, Xinfeng & Ji, Xiaoyu & Yan, Chen & Li, Chaohao & Li, Yichen & Zhang, Zhenning & Xu, Wenyuan. (2023). Learning Normality is Enough: A Software-based Mitigation against Inaudible Voice Attacks.
2. Michelucci, Umberto. (2022). An Introduction to Autoencoders. 10.48550/arXiv.2201.03898.
3. Hadi Abdullah, Kevin Warren, Vincent Bindschaedler, Nicolas Papernot, and Patrick Traynor. Sok: The faults in our asrs: An overview of attacks against automatic speech recognition and speaker identification systems. In 2021 IEEE symposium on security and privacy (SP), pages 730–747. IEEE, 2021.
4. Guoming Zhang, Chen Yan, Xiaoyu Ji, Tianchen Zhang, Taimin Zhang, and Wenyuan Xu. Dolphinattack: Inaudible voice commands. In Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security, pages 103–117, 2017.

THANK YOU